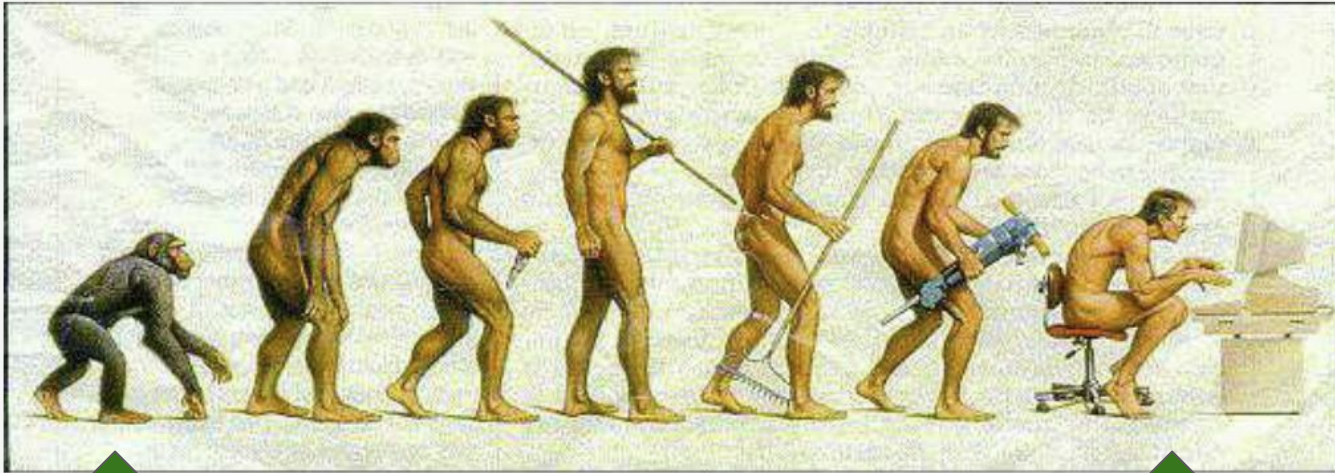


Exploration of Chimpanzee isoform

2026 hackathon Group3



Where we
started

Where we
now

Xinchang Zheng
Numrah Fadra
Md Shakhaowat Hossain
Muteeba Azhar
Nilabja Bhattacharjee
Rojina Sapkota
Umran Yaman
Amit Shenoy
Hannah Chang
Aisha Yousaf
Rajarshi Mondal

Population-scale isoform diversity exploration in chimpanzee

Reference assembly (all work anchored here)

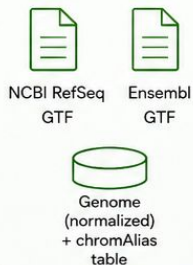
NHGRI_mPanTro3-v2.1_pri (GCF_028858775.2 / GCA_028858775.3)

All sequence names normalized to Ensembl-style (chromAlias table)

GOAL 1 — Chimpanzee annotation integration

Merge major annotations and quantify agreement

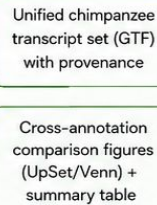
Input



Process

- Normalize sequence names
- Merge with isomatch
- Summarize agreements and disagreements

Output



GOAL 2 — Read-level, population-scale isoform index

Build index from public long-read RNA-seq for frequency queries

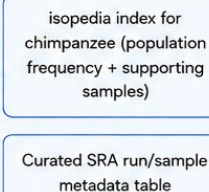
Input



Process

- Collect run list and metadata
- Align reads to genome
- Build isopedia index (per-sample provenance)
- Validate with Goal 1 transcripts

Output

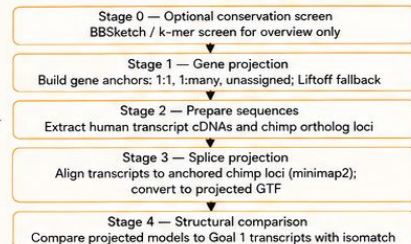
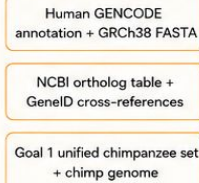


Goal 3 depends on Goal 1 output

GOAL 3 — Human-chimpanzee transcript correspondence (gene-anchored)

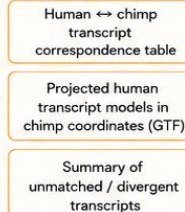
Establish transcript-level correspondence between the human reference annotation and the unified chimpanzee transcript set

Input



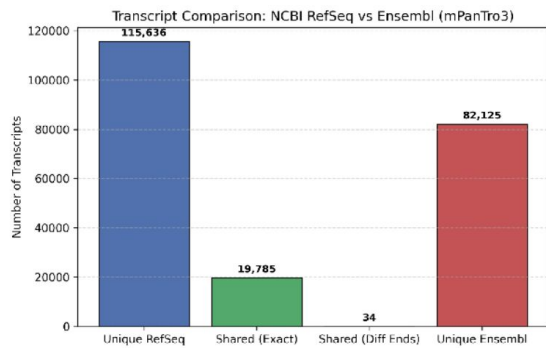
- Match classes**
- exact chain
 - same chain, different ends
 - compatible/contained
 - divergent
 - unalignable
 - no anchor

Output



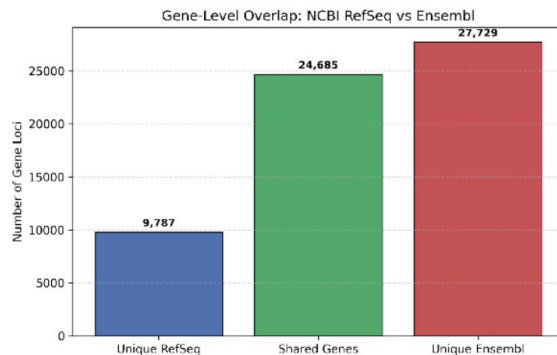
Transcript level comparison

Transcript annotation metric	Count
Total RefSeq transcripts	135,455
— Multi-exon	128,338
— Mono-exon	7,117
Total Ensembl transcripts	101,944
— Multi-exon	77,218
— Mono-exon	24,726
Shared transcripts	19,819
— Exact splice & boundary match	19,785
— Structural match / boundary difference	34
Unique to RefSeq	115,636
Unique to Ensembl	82,125



Gene level comparison

Gene annotation metric	Count
Total RefSeq genes	34,472
Total Ensembl genes	52,414
Shared genes (overlapping loci)	24,685
Unique to RefSeq	9,787
Unique to Ensembl	27,729



What causes the difference?

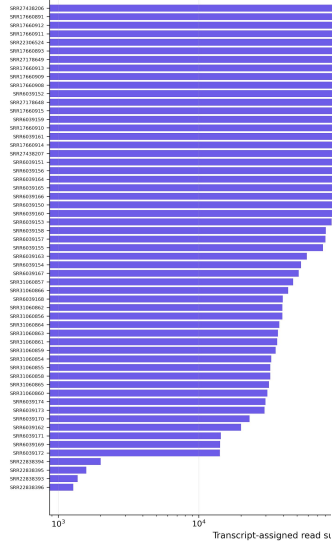
Biotype of unique Ensembl genes	Count	Biotype of unique RefSeq genes	Count
lncRNA	18,557	lncRNA	6,155
misc_RNA	5,384	protein_coding	2,834
protein_coding	1,694	tRNA	505
Y_RNA	751	miRNA	173
snRNA	443	transcribed_pseudogene	43
miRNA	432	V_segment	26
snoRNA	224	misc_RNA	24
IG_V_gene	60	rRNA	20
TR_V_gene	58	C_region	4
pseudogene	48	snRNA	3
rRNA	33		
TR_J_gene	24		
vault_RNA	9		
processed_pseudogene	6		
ribozyme	4		
TR_C_gene	2		

The unified transcript set

`merged_annotation.gtf.merged.gtf.gz`
`merged_annotation.gtf.track.tsv.gz`

Transcript found evidence in each sample

Figure 4. Isopedia expression support per sample



Strong correlation between population frequency and expression level

Figure 9. Read support versus population prevalence

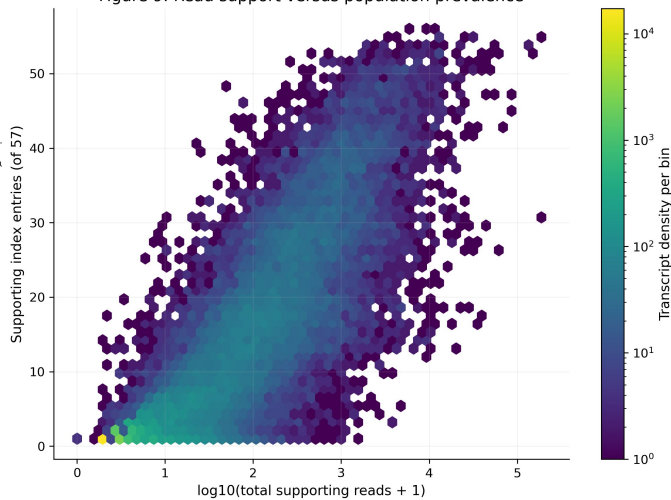
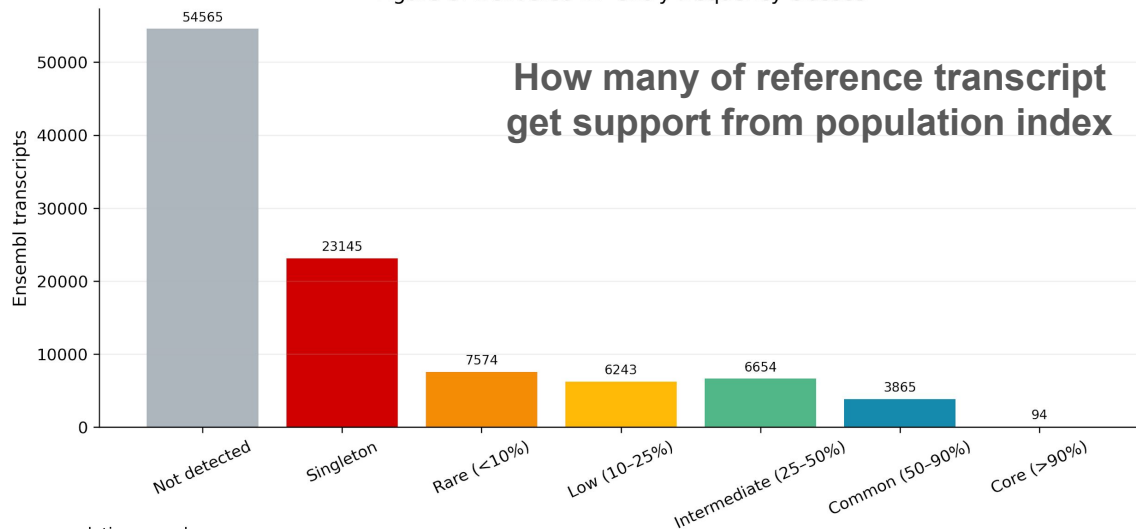
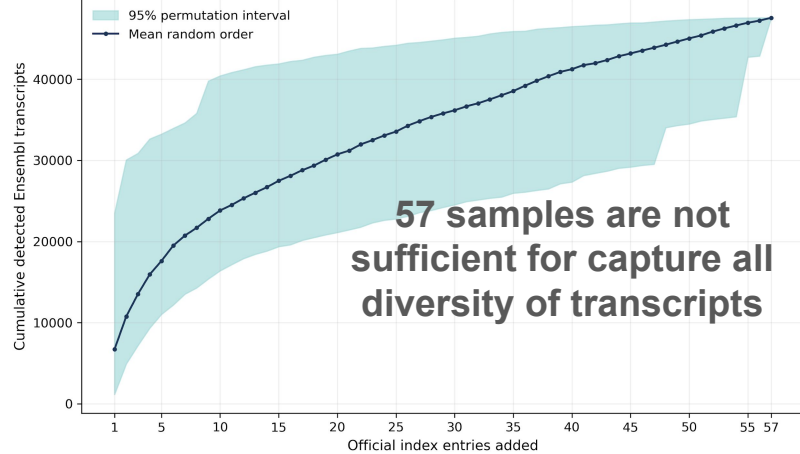


Figure 8. Delivered 57-entry frequency classes



How many of reference transcript get support from population index

Figure 13. Delivered 57-entry discovery curve



57 samples are not sufficient for capture all diversity of transcripts

- **78,733 total human genes** in GENCODE v50.
- **20,107 protein-coding genes** in GENCODE v50.
- **22.6% of protein-coding genes** were linked to an Entrez GeneID using GENCODE metadata and NCBI Ensembl cross-references.
- NCBI chimpanzee RefSeq ortholog assignments (**GCF_028858775.2, RS_2026_05**) provided **17,903 human–chimpanzee ortholog pairs**.
- All identified ortholog pairs were **strictly one-to-one**.
- Overall, **17,869 of 20,107 protein-coding genes (88.9%)** were anchored to a unique chimpanzee locus.

MANE select and protein coding transcripts are better connected between human and chimpanzee

